



A multi-objective evolutionary algorithm for robust positive-unlabeled learning

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ABSTRACT

Positive and unlabeled (PU) learning is to learn a binary classifier with good generalization ability from PU data. A variety of PU learning algorithms with promising performance have been proposed. However, most of them assume that PU samples are “clean”, which is not true in real applications due to the existing noisy or redundant samples. Thus, how to obtain a robust PU classifier with better performance is a challenging problem. To this end, we propose a novel multi-objective evolutionary algorithm to tackle it, named BPUSS-MOEA. Specifically, we firstly transform the robust PU learning into a bi-objective PU sample selection (BPUSS) problem, in which two objectives are designed. One is the number of selected “clean” PU samples and the other is the PU accuracy. Then, a dual-coding scheme is designed to represent the selected “clean” PU samples and the labels of U samples. With the dual-coding scheme, a novel offspring generation strategy is developed to achieve the offsprings with high quality. To further improve the performance of BPUSS-MOEA, an effective population initialization strategy is designed. Experiments on 10 datasets with different noise levels show that compared with the state-of-the-arts, the proposed algorithm demonstrates its robustness in terms of the PU accuracy.

1. Introduction

Positive and unlabeled (PU) learning is an emerging topic in machine learning, whose aim is to train a binary classifier from PU data [26]. In PU learning, the training data is composed of Positive (P) samples and a larger number of Unlabeled (U) samples, where the U samples mean that the labels of samples may be positive or negative. Therefore, the label uncertainty brings the new challenge for the existing supervised learning algorithms. For example, in drug-drug interaction (DDI) predictions, only a few DDIs can be identified as the P samples via drug trials while a large number of DDIs are treated as the U samples [42]. Due to the uncertainty in the labels of U samples, it is difficult to train a binary classifier on the P and U samples to predict the interactions of drug-drug.

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To tackle this issue, many efforts have been made to develop PU learning algorithms, which can be mainly divided into three categories based on how to process the unlabeled data. The first category is to use the two-step approach to construct PU classifiers [3,15,16,19]. Its basic idea is to choose the “reliable” samples from U data, and regard them as the negative samples. Then, with the P samples and the regarded negative samples, different supervised learning algorithms are designed to train binary classifiers for classification. The second category is to treat all U samples as the negative ones and convert PU learning into a positive and negative (PN) learning problem. Since the negative samples may include the mislabeled positive samples [12–14,27], thus, the different strategies are designed to identify the mislabeled positive samples in the converted negative samples. The third category is to regard each U sample as the positive or negative class and transform PU learning into a cost-sensitive learning problem [4,9,10,37]. This category focuses on the classification error cost and improves recall as much as possible. Since the possible positive and negative samples are both considered in the third category, the obtained PU classifiers have shown the superior classification performance.

The works mentioned above (especially the third category) have shown their competitiveness in PU learning. However, in real PU applications, the existing P samples may be contaminated by possible label noises while the U samples may contain redundant samples, which may degrade the performance of PU classifiers. However, most of the existing PU algorithms do not take account of the issues of noisy and redundant samples. As we know, in the diagnosis of cancer, there are likely to be many false positives among the cancer patients, which are called the noisy P samples. While for a large number of U samples (i.e., their diagnosis results are uncertain), there may exist some redundant information, which are useless for classification [41].

Thus, in this paper, we focus on how to train a robust PU classifier from the noisy and redundant PU data. An intuitive idea is to select the “clean” samples from noisy and redundant PU data, with which to train the PU classifier for classification. To implement this idea, in this paper, we propose a novel multi-objective evolutionary algorithm (MOEA) for robust PU learning. The main contributions of this paper can be summarized as follows:

- 1) We firstly transform the robust PU learning into a PU sample selection problem, where the “clean” PU samples are selected for training the PU classifier. Specifically, the robust PU learning is transformed into a bi-objective PU sample selection (BPUSS) problem, whose two objectives are the number of selected PU samples and the classification accuracy of PU classifier.
- 2) To solve the BPUSS problem, a novel MOEA, named BPUSS-MOEA, is proposed. In BPUSS-MOEA, a dual-coding scheme with two layers is firstly designed. The first layer is the binary coding used to denote the selected “clean” samples. The second layer is the real-valued coding used to distinguish the labels of U samples and the selection probabilities of PU samples. Based on the suggested encoding scheme, a dual-coding based offspring generation strategy is designed, with which the proposed algorithm could obtain the PU classifier with high quality. To further enhance the performance of BPUSS-MOEA, an effective population initializing strategy is also developed.
- 3) The effectiveness of the proposed BPUSS-MOEA is verified on 10 real world datasets with different noise levels. Experimental results in comparison with four state-of-the-art PU algorithms show that the proposed algorithm can achieve the PU classifier with high quality in terms of the classification accuracy.

The rest of this paper is organized as follows. Section 2 gives the preliminaries about PU learning and the related works. The proposed multi-objective evolutionary algorithm is illustrated in Section 3. The experimental results are reported in Section 4 by comparing it with four state-of-the-art PU algorithms. Finally, the conclusions and the future works are discussed in Section 5.

2. Preliminaries and related work

In this section, we firstly present some preliminaries about PU learning, then give the related work on PU learning and multi-objective evolutionary optimization, which are closely related to the topic of this paper.

2.1. Preliminaries

PU learning refers to learn from the PU data, with the aim of training a binary classifier to perform the classification tasks. Formally, PU learning can be defined as follows [1]. Given a dataset D , each sample in D can be represented as a triple group (x_i, y_i, s_i) , where $x_i \in \mathbb{R}^l$ denotes the i -th sample with l dimensional real-valued vector, y_i is the label of the sample x_i and s_i is used to denote the sample x_i is positive or unlabeled. Specifically, if $s_i = 1$, the label of the sample x_i is set to 1, i.e., $y_i = 1$, which denotes that x_i is the P sample. If $s_i = 0$, the label of the sample x_i is uncertain, which means that x_i may be positive or negative, i.e., $y_i = 1$ or $y_i = -1$. PU learning is to construct a PU classifier $C(x)$ with good generalization ability, where x represents the sample to be predicted.

2.2. Related work

In recent years, PU learning has attracted a great deal of attention, and the related studies have also increased due to its wide applications in the areas of disease prediction [33], anomaly detection [18] and fake news detection [6], etc. In PU learning, the labels of U samples are uncertain (may be positive or negative), which brings the challenge for supervised learning algorithms. Consequently, a great amount of PU learning algorithms have been proposed and devoted to tackling this challenge.

For example, Liu et al. [16] suggested a two-step approach for PU learning. To be specific, the first step was to identify the “reliable” samples from the U data, which were regarded as the negative ones. The second step was to build the classifier using the P

and the identified negative samples by a supervised learning algorithm. Experimental results have demonstrated the effectiveness of the two-step approach for PU learning. Since the quality of the identified negative samples is key to the performance of the trained classifier, different two-step based approaches have been proposed to select the “reliable” samples. Nguyen et al. [19] suggested a clustering method for PU learning, where the “reliable” samples were extracted by distinguishing the positive and negative boundary of the ground truth. Wang et al. [30] proposed a collaborative representation (CR) based two-step method for PU learning, where the CR technique was designed to identify “reliable” negative examples. Luo et al. [17] proposed a novel PU learning approach, termed as PULNS, in which an effective negative sample selector was designed to select the “reliable” negative samples from U data.

Since the PU learning algorithms mentioned above only utilize the part of U samples to construct PU classifier, the quality of their obtained classifiers still can be further enhanced. To this end, Lee et al. [13] regarded all U samples as the negative ones and converted PU learning into a PN learning problem, where the negative samples may include some mislabeled positive samples. To tackle the issue, a weighted logistic regression model was designed to deal with the mislabeled samples and the experimental results have demonstrated the effectiveness of the suggested approach. Since then, more PU learning algorithms were developed to tackle the converted PN learning problem. Shi et al. [27] suggested to decompose the empirical loss function to model the mislabeled positive samples and estimate the centroid of the negative examples that including the mislabeled positive ones. The experimental results have shown that their decomposition method can effectively alleviate this issue. Gong et al. [12] proposed to convert PU learning into the risk minimization problem in the presence of the mislabeled positive samples, aiming to reduce the adverse impact of such mislabeled samples. Similar research works are also suggested in the literature [14,36]. Since all U samples are considered in the second category approach, the obtained PU classifiers have shown better performance than the first category approach. However, in the U samples, the possible positive samples are ignored, which limits the further performance improvement of the obtained PU classifiers.

Since recognizing the disadvantage of neglecting the possible positive samples in U data, Plessis et al. [9] converted PU learning into a cost-sensitive learning problem, where each U sample was treated as the possible positive or negative sample. To tackle this problem, a novel PU learning algorithm, named uPU, was proposed, where a nonconvex ramp loss function was employed to learn from the transformed PU samples. Experimental results have shown that it could achieve the PU classifier with better performance. From then on, more PU learning algorithms have been proposed to address the converted cost-sensitive learning problem. For example, Plessis et al. [10] further developed different loss functions with convex for PU samples, i.e., a composite convex loss function for P samples and an ordinary convex loss function for U samples. Zhang et al. [37] labeled each U sample as positive and negative class. With the converted samples, a margin-based label disambiguation method was developed for PU learning, namely PULD. In PULD, the true label of each U sample was identified by differentiating the most likely label and the less likely one. Zhao et al. [40] proposed a novel PU learning method (Dist-PU) from the label distribution alignment perspective, where the prior information of the label distribution for unlabeled samples was utilized to obtain the classifier with high quality. More PU algorithms belonging to this category can be found in [5,28,31], and the obtained PU classifiers often showed better performance than the first two category approaches.

Despite that, the approaches mentioned above (including the third category) often require the knowledge of class prior or some assumptions (e.g., smooth and differentiable) on the loss functions to employ the gradient based optimizer [9,10,37]. To this end, Qiu et al. [22] proposed an evolutionary algorithm (EA) for PU learning, named MOEA-PUL, which does not need any prior knowledge or assumptions. In MOEA-PUL, PU learning was firstly modeled as a multi-objective optimization problem, with which the genetic operators were employed to evolve the PU classifier with high quality. Empirical analysis has shown the promising performance compared with the state-of-the-art PU algorithms. Considering the complex PU data in real applications, recently, Qiu et al. [21] further proposed a sparse PU evolutionary algorithm, termed as LMAO-PU, for high-dimensional PU learning. In LMAO-PU, the traditional single-coding scheme was used to distinguish the labels of U samples. Then, a two-stage alternating optimization strategy was designed for obtaining sparse PU classifiers.

These different kinds of PU learning algorithms have shown their competitiveness in obtaining PU classifiers with high quality. However, few of them consider PU learning from the noisy and redundant PU data, which often occurs in the real PU applications. Under this situation, a natural way to tackle the issue is to identify the “clean” samples from the noisy and redundant PU data, and use them for PU learning. However, due to the labels uncertainty of U samples, how to identify the “clean” PU samples is challenging for the existing PU algorithms. To this end, in this paper, we propose a novel MOEA for robust PU learning, in which the identification of “clean” PU samples and the evaluation of PU classifier are modeled as a bi-objective optimization problem. Before elaborating the proposed algorithm, we briefly introduce the preliminary about multi-objective evolutionary optimization, which is adopted to tackle the proposed bi-objective optimization problem.

2.3. Multi-objective evolutionary optimization

In real world, many applications can be modeled as multi-objective optimization problems (MOPs), such as feature selection [24,35], community detection [2,38] and shelter Location [34] etc. Formally, taking the minimization problem as an example, the MOP is defined as follows:

Definition 1. A multi-objective optimization problem (MOP) is defined as:

$$\text{Minimize } \mathbf{F}(\mathbf{X}) = (f_1(\mathbf{X}), f_2(\mathbf{X}), \dots, f_u(\mathbf{X}))^T$$

where $\mathbf{X} = (x_1, x_2, \dots, x_l) \in \Omega$ is the l dimensional decision vector used to represent a solution for the MOP, Ω is the l dimension search space and u is the number of objectives to be optimized. Let X_A and X_B be two solutions of the MOP, if $f_i(\mathbf{X}_A) \leq f_i(\mathbf{X}_B)$, for all $i = 1, 2, \dots, u$, and $\exists j, f_j(\mathbf{X}_A) < f_j(\mathbf{X}_B)$, we call X_A dominate X_B , denoted as $X_A < X_B$. If no decision vector in Ω can dominate the vector X_A , X_A can be called a Pareto optimal solution. All Pareto optimal solutions construct the Pareto set, which is defined as $PS = \{\mathbf{X} \in \Omega | \nexists \mathbf{X}^* \in \Omega, \mathbf{X}^* < \mathbf{X}\}$. The Pareto front is the projection of the Pareto set into the objective space, which is defined as $PF = \{F(\mathbf{X}) | \mathbf{X} \in PS\}$.

During the last decades, evolutionary algorithms have shown the competitiveness in solving the complex optimization problems [11,20]. As one of the important branches in EAs, MOEAs have shown the promising performance in tackling MOPs, since it can achieve a set of trade-off solutions in one run, which can satisfy various requirements in real applications [7,32,39]. In recognizing the superiority of MOEAs, in this paper, we utilize MOEAs to solve the suggested bi-objective optimization problem, which will be illustrated in Section 3.

3. The proposed algorithm

In this section, we firstly transform the robust PU learning into a bi-objective PU samples selection problem, namely BPUSS. Then, the details of the proposed multi-objective evolutionary algorithm, named BPUSS-MOEA, are presented, which includes the suggested dual-coding scheme, the dual-coding based offspring generation strategy, the dual-coding based population initialization strategy, the overall procedure of BPUSS-MOEA and the time complexity analysis.

3.1. The bi-objective PU samples selection problem

In robust PU learning, a key issue is how to select the “clean” samples from the noisy and redundant PU data. To this end, in this paper, we transform the robust PU learning into a bi-objective optimization problem, where the first objective is related to select “clean” samples and the other is to evaluate the performance of PU classifier trained on the selected “clean” PU samples. Specifically, the first objective is measured by the number of selected “clean” PU samples and the other is measured in terms of the classification accuracy of PU classifier. In the suggested bi-objective optimization problem, it is desired to acquire the PU classifier with high quality using as fewer “clean” PU samples as possible. Formally, the suggested bi-objective optimization problem for robust PU learning can be mathematically defined as follows:

$$\begin{cases} \text{Minimize : } f_1(X_{P'U'}) = \frac{|X_{P'U'}|}{|X_{PU}|} \\ \text{Maximize : } f_2(X_{P'U'}) = \frac{\sum_{i=1}^{|X_P|} \sum_{j=1}^{|X_U|} I(\mathbb{C}_{X_{P'U'}}(X_P^i) > \mathbb{C}_{X_{P'U'}}(X_U^j))}{|X_P| \cdot |X_U|} \end{cases} \quad (1)$$

where X_P and X_U respectively denote the original set of P and U samples, X_P' and X_U' respectively represent the set of the selected P and U samples, $X_{PU} = X_P \cup X_U$ and $X_{P'U'} = X_P' \cup X_U'$. X_P^i and X_U^j respectively denote the i -th P sample and the j -th U sample. $\mathbb{C}_{X_{P'U'}}(\cdot)$ denotes the PU classifier trained on the selected PU samples $X_{P'U'}$, which is used to calculate the prediction score of a given sample. $I(\cdot)$ is the indicator function, $I(\cdot) = 1$ if the argument is true, and $I(\cdot) = 0$ otherwise. $|\cdot|$ represents the number of samples for a given sample set.

As shown in Eq. (1), the first objective $f_1(X_{P'U'})$ is to measure the ratio of the selected “clean” PU samples to all PU samples, whose purpose is to select as small number of “clean” PU samples as possible. The second objective $f_2(X_{P'U'})$ is designed to measure the classification performance of the PU classifier. Its basic idea comes from the fact that a good PU classifier should have larger predicted scores on P samples than those on U samples. Thus, $f_2(X_{P'U'})$ is used to calculate the percentage that the prediction scores on the P samples are larger than those on the U samples to all pairwise (positive and unlabeled) samples. With the suggested two objectives, the robust PU learning is modeled as a bi-objective PU samples selection problem. Before giving the details of the proposed method, Table 1 lists the main notations used in this paper for our method.

3.2. The suggested dual-coding scheme

Since MOEAs have shown the promising performance in tackling MOPs [2,24], thus, we design a novel MOEA to solve the suggested bi-objective BPUSS problem. We note that in Eq. (1), there are two issues needed to be tackled. One is to select the “clean” PU samples and the other is to distinguish the labels of U samples, which brings the new difficulty for widely used single-coding schemes (e.g., the binary or real-valued coding) in MOEAs.

To this end, a dual-coding scheme consisting of two layers is proposed. Specifically, the first layer is the binary coding (i.e., the (0-1) coding) for all PU samples, whose aim is to select the “clean” samples. In the binary (0-1) coding, ‘0’ denotes the sample is not selected while ‘1’ denotes the sample is selected. All the selected samples are regarded as the “clean” samples and are further used for training PU classifiers. The second layer is the real-value coding, whose main aim is to distinguish the labels of U samples. Specifically, ‘+’ and ‘-’ denote the label of the given U sample is positive or negative. In addition, the absolute value is used to denote the probability of the given sample (P or U sample) regarded as the “clean” sample. The larger the absolute value, the higher probability of the corresponding sample being regarded as the “clean” sample.

Table 1

The main notations used in this paper.

Notation	Description
X_P, X_U	the original set of P and U samples, $X_{PU} = X_P \cup X_U$
X'_P, X'_U	the set of the selected P and U samples, $X_{PU'} = X'_P \cup X'_U$
X^i_P, X^j_U	the i -th P sample and the j -th U sample
$C_{X_{PU'}}(\cdot)$	the PU classifier trained on the selected PU samples $X_{PU'}$
$I(\cdot)$	the indicator function, $I(\cdot) = 1$ if the argument is true, and $I(\cdot) = 0$ otherwise.
$f_1(X_{PU'})$	the ratio of the selected “clean” PU samples to all PU samples
$f_2(X_{PU'})$	the classification performance of the PU classifier
P^t	the t -th generation population
ind_i^t	the i -th individual with the dual-coding in P^t
$bind_i^t$	the binary coding part of ind_i^t
$rind_i^t$	the real-valued coding part of ind_i^t
$ind_{i,j}^t$	the j -th bit of ind_i^t
$bind_{i,j}^t$	the j -th binary bit in $bind_i^t$
$rind_{i,j}^t$	the j -th real number bit in $rind_i^t$
$ rind_{i,j}^t $	the selection probability of the j -th sample
Rnd	an uniformly distributed random number
$Elite_{real}^t$	the real-valued coding part of the elite individual $Elite^t$ in P^t
$Num(X_{P,j}^{pre})$	the number of the predicted positive samples in the neighbors of the j -th sample
$Num(X_{N,j}^{pre})$	the number of the predicted negative samples in the neighbors of the j -th sample

PU samples	X_P^1	X_P^2	X_P^3	X_P^4	X_U^1	X_U^2	X_U^3	X_U^4	X_U^5	X_U^6
The index number of samples j	1	2	3	4	5	6	7	8	9	10
$bind_i^t$	0	1	1	1	1	0	1	1	1	0
$rind_i^t$	0.6	0.2	0.2	0.4	+0.4	-0.8	+0.4	-0.4	-0.6	-0.6

Fig. 1. An example of the binary coding ($bind_i^t$) and the real-valued coding ($rind_i^t$) of individual (ind_i^t).

Let ind_i^t be the i -th individual with the dual-coding in the t -th generation population P^t , where $bind_i^t$ and $rind_i^t$ are the binary coding part and real-valued coding part of ind_i^t , respectively. Let $bind_i^t = (bind_{i,1}, \dots, bind_{i,m})$ and $rind_i^t = (rind_{i,1}, \dots, rind_{i,m})$, where m is the length of ind_i^t (i.e., the number of the PU samples). For the j -th element in ind_i^t , $ind_{i,j}^t$ is composed of the values of the two-layer coding, i.e., the binary coding value $bind_{i,j}$ and the real-valued coding value $rind_{i,j}$. $bind_{i,j} \in \{0, 1\}$ is used to denote the j -th sample is selected or not, where ‘1’ denotes the sample is selected while ‘0’ is not. $|rind_{i,j}| \in (0, 1)$ is the selection probability of the j -th sample. For the j -th U sample X_U^j , the $sign(rind_{i,j}) \in \{+, -\}$ gives its label, where ‘+’ (‘-’) denotes positive (negative) label. In addition, the larger the absolute value $|rind_{i,j}|$, the higher probability of the corresponding the j -th sample being treated as the “clean” sample.

Fig. 1 illustrates the suggested dual-coding scheme, where ind_i^t is the i -th individual composed of the binary coding $bind_i^t$ and the real-valued coding $rind_i^t$. The length of ind_i^t is the same as the number of PU samples, i.e., $m = 10$, including four P samples $\{X_P^1, \dots, X_P^4\}$ and six U samples $\{X_U^1, \dots, X_U^6\}$ (see the grey and yellow parts in the first line). Specifically, the binary coding $bind_i^t = \{0, 1, 1, 1, 1, 0, 1, 1, 1, 0\}$ means that three P samples $\{X_P^2, X_P^3, X_P^4\}$ and four U samples $\{X_U^1, X_U^3, X_U^4, X_U^5\}$ are selected as the “clean” samples (see the grey and yellow parts in the binary coding part $bind_i^t$). The real-valued coding $rind_i^t$ gives the selection probability of each sample and the predicted label (‘+’ or ‘-’) for each U sample.

Let $|rind_{i,j}^t|$ be the selection probability of the j -th sample, where $j = 1, \dots, 10$, in this figure. $|rind_{i,4}^t| = 0.4$ means that the selection probability of X_P^4 is 0.4, which is painted with gray in the real-valued coding part $rind_i^t$. $|rind_{i,7}^t|$ and $|rind_{i,8}^t|$ are +0.4 and -0.4, which represents the selection probabilities of X_U^3 and X_U^4 are both 0.4 and respectively predicted as the positive (‘+’) sample and negative (‘-’) sample, which are painted with yellow in the real-valued coding part.

With the suggested dual-coding scheme, we can represent both the selected “clean” samples and the labels of U samples in the same individual. However, in MOEAs, the traditional evolutionary operators (e.g., crossover and mutation) used to generate the offsprings are designed for the single-coding scheme, which are not suitable for the suggested dual-coding scheme. To this end, in this paper, we propose a dual-coding based offspring generation strategy, which will be elaborated in the next section.

3.3. The dual-coding based offspring generation strategy

In this section, a dual-coding based offspring generation strategy is suggested to generate new offsprings, which is implemented by two updating stages. Specifically, for the i -th individual in population P^t , i.e., ind_i^t , the first stage is to update the binary encoding

part, i.e., $bind_i^t$, and the second stage is to update the real-valued encoding part, i.e., $rind_i^t$. With the proposed two updating stages, the promising offspring population P^{t+1} are generated. Algorithm 1 further presents the procedure of the suggested dual-coding based offspring generation strategy. In the following, we will discuss them in detail.

Algorithm 1: DualCodingOffspringGeneration.

Require: P^t : the t -th generation population, PS : the set of elite individuals in the population P^t , $popsiz$: the population size, m : the length of the individuals, t : the current number of generation, $Maxiter$: the maximum number of the generation.
Ensure: The offspring population P^{t+1} ;
1: $P' \leftarrow \text{BinaryIndividualUpdate}(P^t, popsiz, Maxiter)$;
//Algorithm 2
2: $P^{t+1} \leftarrow \text{RealValueIndividualUpdate}(P', popsiz, PS, m, t)$; //Algorithm 3

3.3.1. The stage of updating the binary coding part

To update the binary encoding part of each individual, a probability-guided sample selection operator is proposed to generate the new binary offspring, which consists of two steps. The first step is to select the binary bits should be updated. The second step is to update the binary bits selected in the first step.

In the first step, we need to determine how many binary bits in an individual needed to be updated, whose basic idea is from the following intuition. In the early stage of the evolution, the population needs explore more potential regions to find the global optimal solution, while in the late stage of the evolution, the population may be close to the global optimal solution and more exploitation need to be performed around it. Based on this intuition, we hope that in the early stage of the evolution, more binary bits are selected to be updated while in the late stage of the evolution, less binary bits are considered to be updated. To implement this idea, we utilize the Manhattan distance to adaptively adjust the number of binary bits to be updated at each generation. To be specific, let $bind_i^t$ be the first layer binary coding part of individual ind_i^t and we assume the updated binary coding part as $bind_i^{t+1}$. Thus, the number of binary bits that needs to be updated in $bind_i^t$ is denoted as the Manhattan distance between $bind_i^t$ and $bind_i^{t+1}$, which is defined as follows.

$$\text{ManhatDist}(bind_i^t, bind_i^{t+1}) \leq \lceil m \times (1 - \frac{t}{Maxiter}) \rceil \quad (2)$$

where m is the length of individual $bind_i^t$ and $Maxiter$ is the maximum number of generation. $\lceil m \times (1 - \frac{t}{Maxiter}) \rceil$ represents the maximum number of binary bits needed to be updated in $bind_i^t$, which can ensure the balance of the exploration and exploitation ability of the population during the evolutionary. After that, $\lceil m \times (1 - \frac{t}{Maxiter}) \rceil$ binary bits are selected randomly from the $bind_i^t$ and used to perform the sample selection operator suggested in the following.

In the second step, a probability-guided sample selection operator is designed to update the binary bits selected in the first step. The main idea is to utilize the selection probabilities of PU samples in $rind_i^t$ to guide the update of the binary bits selected in the first step. To be specific, for the j -th sample with higher selection probability, if it is not selected in the t -th generation, it will be selected in $(t + 1)$ -th generation with higher probability. On the contrary, for the j -th sample with lower selection probability, although it is selected in the t -th generation, it will be deleted with higher probability in $(t + 1)$ -th generation. Thus, the suggested selection probability-guided sample selection operator is defined as follows.

$$bind_{i,j}^{t+1} = \begin{cases} 1 - bind_{i,j}^t, & bind_{i,j}^t = 1 \wedge |rind_{i,j}^t| < Rnd \quad \text{or} \\ & bind_{i,j}^t = 0 \wedge |rind_{i,j}^t| \geq Rnd \\ bind_{i,j}^t, & bind_{i,j}^t = 1 \wedge |rind_{i,j}^t| \geq Rnd \quad \text{or} \\ & bind_{i,j}^t = 0 \wedge |rind_{i,j}^t| < Rnd \end{cases} \quad (3)$$

where $bind_{i,j}^t$ is the binary value of the j -th bit in $bind_i^t$, $|rind_{i,j}^t|$ is the real number value of the j -th bit in $rind_i^t$ and $Rnd \in (0, 1)$ is an uniformly distributed random number. As can be seen from Eq. (3), if the j -th sample with higher probability has not been selected, that is, $bind_{i,j}^t = 0$, the j -th sample will be selected in the next generation, that is, $bind_{i,j}^{t+1} = 1$. On the contrary, if the j -th sample with lower selection probability has been selected, that is, $bind_{i,j}^t = 1$, the j -th sample will be deleted from the next generation, that is, $bind_{i,j}^{t+1} = 0$. It can guarantee that the sample with larger selection probability can be selected as much as possible. Algorithm 2 gives the details of suggested binary coding part updating.

Fig. 2 gives an example to illustrate the suggested binary coding updating. Let $bind_i^t = \{0, 1, 1, 1, 1, 0, 1, 1, 1, 0\}$ be the binary coding part of ind_i^t , in which three P samples $\{X_P^2, X_P^3, X_P^4\}$ and four U samples $\{X_U^1, X_U^3, X_U^4, X_U^5\}$ are selected and painted with yellow (see the left part of this figure). $rind_i^t = \{0.6, 0.2, 0.2, 0.4, +0.4, -0.8, +0.4, -0.4, -0.6, -0.6\}$, where $|rind_i^t|$ gives the selection probability of each sample. For the six U samples, $\{X_U^1, X_U^3\}$ and $\{X_U^2, X_U^4, X_U^5, X_U^6\}$ are respectively labeled as the positive (denoted as '+') and negative (denoted as '-') samples. Assuming the maximum number of generation $Maxiter$ is 100, the current number of generation t is 70 and the length of the individual m is 10, $\lceil m \times (1 - \frac{t}{Maxiter}) \rceil$ equals to 3, which means there are three binary bits to be selected and updated. In this example, we assume that $\{X_P^2, X_P^3, X_U^6\}$ (marked in red in the left part) are randomly

Algorithm 2: BinaryIndividualUpdate.

Require: P^t : the t -th generation population, $popsize$: the population size, $Maxiter$: the maximum number of generation.
Ensure: P' : the population that the updated binary coding part;
1: **for** $i = 1$ to $popsize$ **do**
2: $bind_i^t \leftarrow$ get the binary coding part of ind_i^t ;
3: $rind_i^t \leftarrow$ get the real-valued coding part of ind_i^t ;
4: $bind_i^{t+1} \leftarrow bind_i^t$;
5: $Num_bin \leftarrow$ determine the number of binary bits according to Eq. (2);
6: $index \leftarrow$ randomly select Num_bin bits from $bind_i^t$ to be updated;
7: **for each** j in $index$ **do**
8: **if** $bind_{i,j}^t = 1 \wedge |rind_{i,j}^t| < Rnd$ or $bind_{i,j}^t = 0 \wedge |rind_{i,j}^t| \geq Rnd$ **then**
9: $bind_{i,j}^{t+1} = 1 - bind_{i,j}^t$;
10: **end if**
11: **end for**
12: $P' \leftarrow \{bind_i^{t+1}, rind_i^t\}$;
13: **end for**

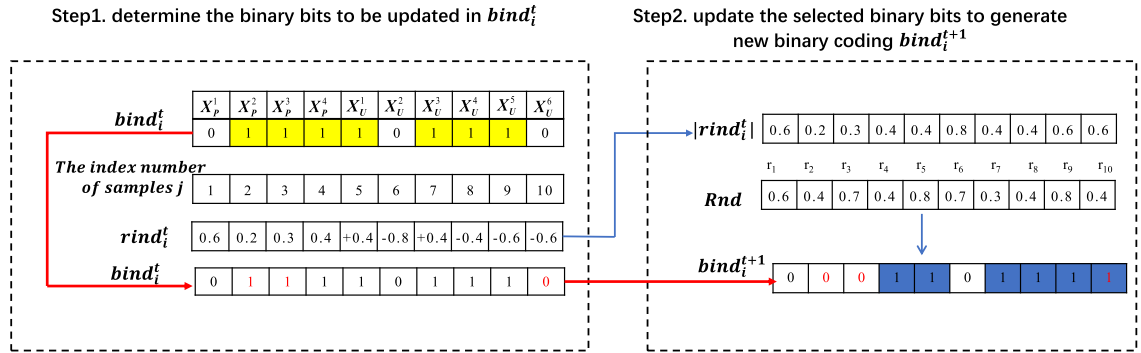


Fig. 2. An illustrative example of the suggested binary coding updating.

selected to perform the following updating operator. As shown in this figure, $bind_{i,2}^t = 1$ and $bind_{i,3}^t = 1$, which means the sample X_P^2 and X_P^3 are selected in P^t . $rind_{i,2}^t = 0.2$ and $rind_{i,3}^t = 0.3$ denote the selection probabilities of X_P^2 and X_P^3 are 0.2 and 0.3, which are respectively smaller than the random numbers 0.4 and 0.7. Thus, in the updated binary offspring coding part $bind_i^{t+1}$, $bind_{i,2}^{t+1}$ and $bind_{i,3}^{t+1}$ are respectively updated as 0, which means that the samples X_P^2 and X_P^3 are deleted from the set of the selected samples. For the unlabeled sample X_U^6 , its selection probability $|rind_{i,10}^t| = 0.6$, which is larger than a generated random number 0.4. Therefore, the value of $bind_{i,10}^{t+1}$ will be updated as 1, that is the sample X_U^6 is added into the set of the selected samples. Finally, in the updated binary offspring coding part $bind_i^{t+1}$, $\{X_P^4, X_U^1, X_U^3, X_U^4, X_U^5, X_U^6\}$ are the new selected samples from the original PU data and painted with blue in $bind_i^{t+1}$ (see the right part of this figure).

3.3.2. The stage of updating the real-valued coding part

In this section, an elite individual based update operator is proposed to update the real-value coding part of $rind_i^t$. The main procedure is composed of two steps. The first step is to utilize the t -th generation elite individuals to determine the labels of U samples in the $(t+1)$ -th generation, i.e., $sign(rind_i^{t+1})$. The second step is to determine the selection probabilities of the samples in the $(t+1)$ -th generation, i.e., $|rind_i^{t+1}|$.

The basic idea of the first step is that the individuals in PF (called the elite individuals) usually have better qualities than the other individuals in population. Therefore, we assume $Elite_{real}^t$ as the real-valued coding part of the elite individual $Elite^t$. The difference of between $rind_i^t$ and $Elite_{real}^t$ to determine the labels of U samples in $rind_i^{t+1}$. The larger the difference is, the higher probability of the label of the U sample is changed. To be specific, given the $rind_i^t$ and $Elite_{real}^t$, a sigmoid function $H(\cdot)$ is designed to measure the difference and map them into a set of label updating probabilities, which is defined as follows:

$$H = \frac{1}{1 + e^{-10(|Elite_{real}^t| - |rind_i^t|)^2}} \quad (4)$$

where $Elite_{real}^t$ is the real-valued coding of the elite individual $Elite^t$ selected randomly from the set of Pareto solution (PS), $rind_i^t$ is the real-value coding part of ind_i^t . It is noted that the process of determining the labels suggested above is only applied to the U samples. Finally, we determine the labels of U samples in $rind_i^{t+1}$ by comparing the obtained label updating probabilities to the generated random numbers.

The second step is to update the selection probabilities of the samples selected in $bind_i^{t+1}$. The basic idea is from the following intuition. The classification accuracy of the PU classifier is determined by the qualities of the selected “clean” PU samples. The

Algorithm 3: RealValueIndividualUpdate.

Require: P' : the population that the updated binary coding part, $popsiz$: the population size, PS : the set of Pareto solutions, m : the length of the individuals, t : the number of generation.

Ensure: The offspring population P^{t+1} ;

- 1: **for** $i = 1$ to $popsiz$ **do**
- 2: $bind_i^{t+1} \leftarrow$ get the binary coding part of the i -th individual in P' ;
- 3: $rind_i^t \leftarrow$ get the real-valued coding part of the i -th individual in P' ;
- 4: $sign(rind_i^t) \leftarrow$ get the labels of the U samples from the $rind_i^t$;
- 5: $Elite^t \leftarrow$ select an elite individual randomly from PS ;
- 6: $Elite_{real}^t \leftarrow$ get the real-valued coding part of the elite individual $Elite^t$;
- 7: $sign(Elite_{real}^t) \leftarrow$ get the labels of the U samples from the $Elite_{real}^t$;
- 8: $rind_i^{t+1} \leftarrow rind_i^t$;
- 9: **Step 1: Determine the labels of unlabeled samples**
- 10: $index \leftarrow$ find the different bits between $sign(Elite_{real}^t)$ and $sign(rind_i^t)$
- 11: **for each** j in $index$ **do**
- 12: $H_j \leftarrow$ Calculate the label updating probability value according to Eq. (4);
- 13: **if** $H_j > random()$ **then**
- 14: $sign(rind_{i,j}^{t+1}) = \sim(sign(rind_{i,j}^t))$;
- 15: **end if**
- 16: **end for**
- 17: $index \leftarrow \emptyset$;
- 18: **Step 2: Update the selection probabilities of the samples selected in $bind_i^{t+1}$**
- 19: $index \leftarrow$ find the indexes of the samples that selected in $bind_i^{t+1}$;
- 20: $X_{P'U'} \leftarrow$ select the samples from $X_{P'U'}$ with $index$;
- 21: $f_1(X_{P'U'}) \leftarrow$ the PU accuracy of the PU classifier trained on the $X_{P'U'}$;
- 22: **for each** j in $index$ **do**
- 23: $|rind_{i,j}^{t+1}| \leftarrow$ using $|rind_{i,j}^t|$, $f_1(X_{P'U'})$ and $|Elite_{real}^t|$ according to Eq. (5);
- 24: **end for**
- 25: $P^{t+1} \leftarrow \{bind_i^{t+1}, rind_i^{t+1}\}$;
- 26: **end for**

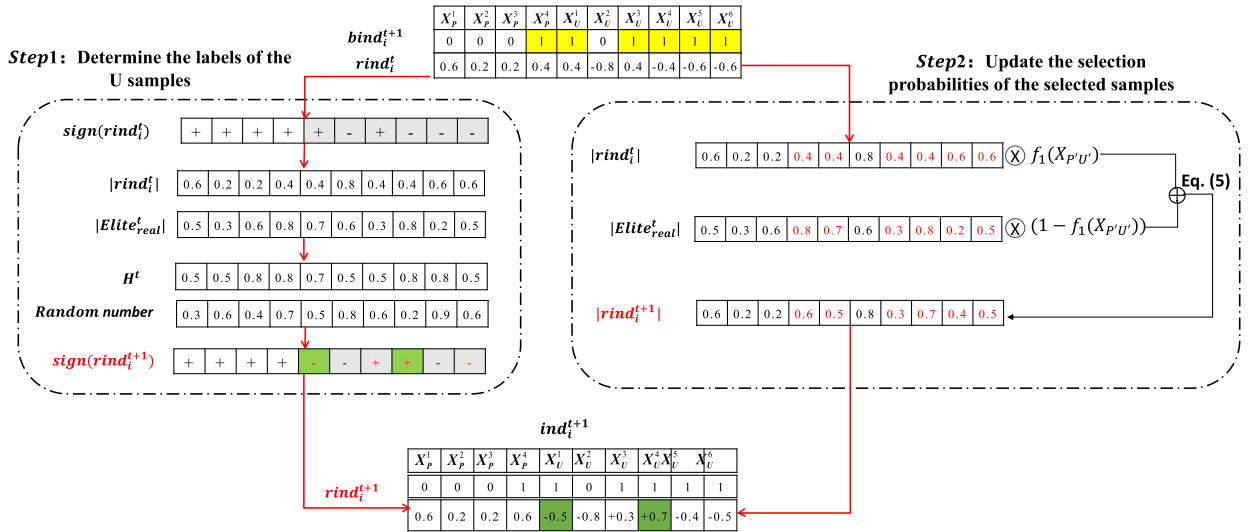


Fig. 3. An illustration of the suggested real-valued coding updating stage.

higher classification accuracy of the PU classifier, the higher quality of the selected PU samples. Based on this idea, the classification accuracy is used as the weight to adaptively update the selection probabilities of the samples selected in $bind_i^{t+1}$. Given $rind_i^t$ and $Elite_{real}^t$, the suggested elite individual based updating operator is defined as follows:

$$|rind_i^{t+1}| = |rind_i^t| * f_1(X_{P'U'}) + Elite_{real}^t * (1 - f_1(X_{P'U'})) \quad (5)$$

where $f_1(X_{P'U'})$ (see Eq. (1)) is used to measure the classification accuracy of the PU classifier trained on the selected PU samples $X_{P'U'}$. As can be seen from Eq. (5), if $f_1(X_{P'U'})$ is low, it means that the more noisy and redundant samples may be included in $X_{P'U'}$. Thus, the probabilities of the selected PU samples in $bind_i^{t+1}$ are updated by learning from the elite individual $Elite_{real}^t$ with the larger weight of $(1 - f_1(X_{P'U'}))$. Algorithm 3 gives the process of the suggested real-valued coding part updating.

Fig. 3 gives an example to illustrate the process of updating the real-valued coding part, where $rind_i^t$ and $rind_i^{t+1}$ are the real-valued coding part of the parent individual ind_i^t and its offspring individual ind_i^{t+1} . The gray part of the $sign(rind_i^t)$ presents the

labels of the six U samples, where $\{X_U^1, X_U^3\}$ and $\{X_U^2, X_U^4, X_U^5, X_U^6\}$ are predicted as positive and negative samples, respectively. The whole updating process includes two steps. The first step is to determine the labels of the U samples (shown in Step 1). In Step 1, the difference between $|rind_i^t|$ and $|Elite_{real}^t|$ is mapped into a set of label updating probabilities (see H^t). The labels of the U samples in the $rind_i^{t+1}$ are determined by comparing the obtained H^t with a set of random numbers. As shown in this example, the labels of unlabeled sample $\{X_U^1, X_U^4\}$ are updated as '-' and '+' respectively, which is shown in green part of $sign(rind_i^{t+1})$. The second step is to update the selection probabilities of the selected samples in $bind_i^{t+1}$ (marked in red and shown in Step 2), i.e., $\{X_P^4, X_U^1, X_U^3, X_U^4, X_U^5, X_U^6\}$, where $f_1(X_{P'U'})$ is used as a weight for updating the selection probabilities of the selected PU samples according to Eq. (5).

3.4. The dual-coding based population initialization strategy

The dual-coding based offspring generation strategy suggested above is used to generate the promising offspring individuals. To further enhance the performance of BPUS-MOEA, a dual-coding based population initialization strategy is proposed, which is composed of two steps. The first step is to randomly initialize the binary coding part of the individual, which is used to determine which samples are selected for classification. The second step is to utilize the neighbor samples of each sample to initialize the real-valued coding part of the individual. With the two steps, the initial population with promising performance is generated and the process of initialization is presented in Algorithm 4.

Algorithm 4: DualCodingBasedInitialization.

Require: X_P : the set of P samples, X_U : the set of U samples, $popsize$: the population size, L : the size of neighbors;
Ensure: P : the initial population;
1: $m \leftarrow |X_P| + |X_U|$ // the length of each individual;
2: **for** $i = 1$ to $popsize$ **do**
3: $bind_i \leftarrow random(0, 1, m)$; // initialize m binary bits randomly for $bind_i$
4: $X_{P'U'} \leftarrow$ generate the training data with the selected samples according to $bind_i$
5: $C_{kNN}(X_{P'U'}) = kNN(X_{P'U'})$; //construct the classifier $C_{kNN}(X_{P'U'})$
6: $rind_i \leftarrow zeros(1, m)$; // initialize m zeroes for $rind_i$
7: **for** $j = 1$ to $|X_P|$ **do**
8: $Num(X_{P,j}^{Pre}) \leftarrow$ the number of predicted positive samples obtained by $C_{kNN}(X_{P'U'})$ in the L neighbors of the X_P^j ;
9: $rind_{i,j} = \frac{Num(X_{P,j}^{Pre})}{L}$;
10: **end for**
11: **for** $j = |X_P| + 1$ to m **do**
12: $Num(X_{P,j}^{Pre}) \leftarrow$ the number of predicted positive samples obtained by $C_{kNN}(X_{P'U'})$ in the L neighbors of X_U^j ;
13: $Num(X_{N,j}^{Pre}) \leftarrow$ the number of predicted negative samples obtained by $C_{kNN}(X_{P'U'})$ in the L neighbors of X_U^j ;
14: $|rind_{i,j}| = \frac{Num(X_{N,j}^{Pre})}{L}$;
15: **if** $Num(X_{P,j}^{Pre}) > Num(X_{N,j}^{Pre})$ **then**
16: $rind_{i,j} = |rind_{i,j}| * (+1)$;
17: **else**
18: $rind_{i,j} = |rind_{i,j}| * (-1)$;
19: **end if**
20: **end for**
21: $P \leftarrow \{bind_i, rind_i\}$;
22: **end for**

We take the individual ind_i as an example to illustrate the suggested population initialization strategy. The first step is to generate randomly the binary coding part of the ind_i , i.e., $bind_i$, where each bit is set to 0 or 1 randomly. With the selected samples (denoted as '1'), a classifier C_{kNN} is constructed by using a k -Nearest Neighbors (kNN) algorithm, where the selected U samples are simply treated as negative samples (see Lines 3 - 5 in Algorithm 4). The second step is to generate the real-valued coding part of ind_i , i.e., $rind_i$. The basic idea is to utilize the predicted labels of most of the neighbors of each sample to determine the value of each bit in $rind_i$, where the predicted labels are obtained by the C_{kNN} classifier. To be specific, for the j -th sample, in its L neighbor samples, the more samples are predicted as positive (negative), the more probability of the j -th sample is regarded as the positive (negative) sample. Let $Num(X_{P,j}^{Pre})$ and $Num(X_{N,j}^{Pre})$ respectively be the number of the predicted positive samples and predicted negative samples in the L neighbor samples of the j -th sample, where $Num(X_{P,j}^{Pre}) + Num(X_{N,j}^{Pre}) = L$. Therefore, for the j -th bit in $rind_i$, its value is initialized as $\frac{Num(X_{P,j}^{Pre})}{L}$ (if the sample is positive) or $\frac{Num(X_{N,j}^{Pre})}{L}$ (if the samples are unlabeled). Additionally, due to the labels uncertainty of U samples, the initial labels of U samples are also presented based on the same idea mentioned above. To be specific, for the j -th U sample, in its L neighbor samples, if $Num(X_{P,j}^{Pre}) < Num(X_{N,j}^{Pre})$, the label of X_U^j is initialized as negative and the value of the j -th bit in $rind_i$ is multiplied by '-1', otherwise is positive and the value of the j -th bit is multiplied by '+1' (see Lines 7 - 20 in Algorithm 4).

Fig. 4 presents an example for generating the initial individual ind_i with the suggested initialization strategy. The first step is for initial binary coding part $bind_i$ (see Fig. 4 (a)). The second step is for initial real-valued coding part $rind_i$ (see Fig. 4 (b) - (d)). Specifically, Fig. 4 (a) gives the PU dataset X_{PU} including four P samples $\{X_P^1, \dots, X_P^4\}$ and six U samples $\{X_U^1, \dots, X_U^6\}$. We

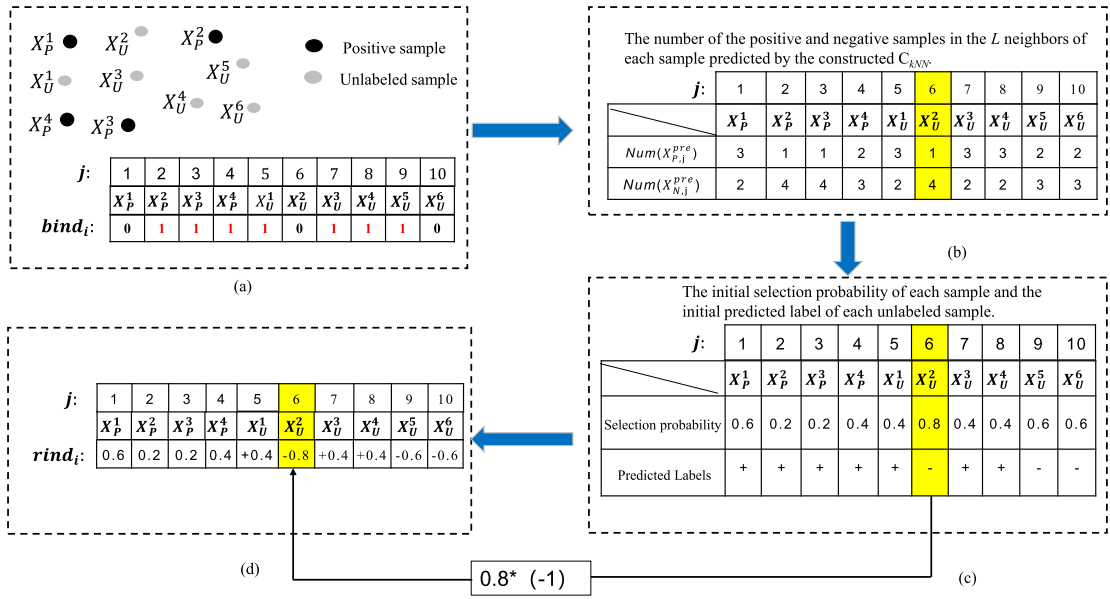


Fig. 4. An illustration of the suggested dual-coding based population initialization strategy. (a) is for the initial binary coding part (the first step); (b)-(d) are for the initial real-valued coding part (the second step).

randomly generate the initial binary coding part $bind_i = \{0, 1, 1, 1, 1, 0, 1, 1, 1, 0\}$, where $\{X_P^2, X_P^3, X_P^4, X_U^1, X_U^3, X_U^4, X_U^5\}$ are selected and marked in red color. For the j -th sample in X_{PU} , Fig. 4 (b) gives the number of the predicted positive and negative samples in its L neighbors, i.e., $Num(X_{P,j}^{Pre})$ and $Num(X_{N,j}^{Pre})$, where L is assumed to be 5 and j is numbered from 1 to 10 used to denote ten PU samples. Taking the sample X_U^2 as an example ($j = 6$), in its 5 neighbor samples, there are one sample is predicted as positive and four samples are predicted as negative, that is $Num(X_{P,6}^{Pre}) = 1$ and $Num(X_{N,6}^{Pre}) = 4$ painted with yellow in Fig. 4 (b). Fig. 4 (c) presents the initial value of each bit in $rind_i$. We take $rind_{i,6}$, which corresponds the U sample X_U^2 , as an example to describe the details. In the five neighbor samples of the X_U^2 , one is predicted as positive and the other four ones are predicted negative. Thus, the selection probability of X_U^2 is equal to $4/5 = 0.8$, i.e., $|rind_{i,6}| = 0.8$. In addition, due to $Num(X_{P,6}^{Pre}) < Num(X_{N,6}^{Pre})$, the initial label of X_U^2 is set to '-'. Fig. 4 (d) gives the initial real-valued coding part $rind_i$. We still take the $rind_{i,6}$ as an example, since the selection probability of $rind_{i,6}$ is 0.8 and predicted as the negative, the value of $rind_{i,6}$ is initialized as $0.8 * (-1) = -0.8$.

Algorithm 5: The framework of BPUS-MOEA.

Require: X_P : the set of positive samples, X_U : the set of unlabeled samples, m : the length of the individuals, $Maxiter$: the maximum number of the generation, $popsiz$: the size of the population, L : the size of the neighbors;
Ensure: The final set of non-dominated individuals FS ;
Step1: Generate the promising initial population
 1: $[P^1] \leftarrow \text{DualCodingBasedInitialization}(X_P, X_U, popsiz, L)$; //Algorithm 4
 2: $PS \leftarrow \text{SelectNonDominatedSet}(P^1)$;
 // PS is the non-dominated set of the initial population P^1 .
Step2: Evolve the population to generate the competitive offspring population
 3: $t = 1$;
 4: **while** $t \leq Maxiter$ **do**
 5: $P^t \leftarrow \text{MatingSelection}(P^t, popsiz)$;
 6: $O \leftarrow \text{DualCodingOffspringGeneration}(P^t, PS, popsiz, m, t, Maxiter)$; //Algorithm 1
 7: $[P^{t+1}, PS] \leftarrow \text{EnvironmentalSelection}(P^t \cup O, popsiz)$;
 8: $t = t + 1$;
 9: **end while**
Step3: Obtain the non-dominated individuals as the final output
 10: $FS \leftarrow \text{SelectNonDominatedSet}(P^t)$; //Select the non-dominated individuals from P^t .

3.5. The overall procedure of BPUS-MOEA

In this paper, NSGA-II [8] is employed as the basic optimization framework for the suggested BPUS problem, which is composed of three steps and shown in Algorithm 5. In the first step, we utilize the suggested dual-coding based population initialization strategy (Algorithm 4) to generate the initial population with promising performance (see Line 1). The second step is to evolve the population

Table 2
10 datasets with different characteristics.

Dataset	#Samples	#Features	Num of P	Num of N
Sonar	208	60	97	111
Heart	270	13	120	150
Balance	625	4	288	337
Breast	683	10	239	444
Transfusion	748	4	178	570
Diabetes	768	8	500	268
Vehicle	846	18	212	634
German	1000	24	300	700
Nba	1340	19	831	509
Madelon	2600	500	1300	1300

to obtain the competitive offspring population until the termination condition is satisfied (see Lines 3 to 9). Firstly, the binary tournament selection is utilized to select the parents to perform the mating operation in *MatingSelection()*. Secondly, we utilize the suggested dual-coding based offspring generation strategy (Algorithm 1) to generate the competitive individuals as the offsprings (Line 6). Thirdly, the non-dominated sorting and crowded distance in NSGA-II are employed for the environmental selection in *EnvironmentalSelection()*. In the third step, the non-dominated sort method in NSGA-II is adopted to obtain the non-dominated individuals from the final population (see Line 10).

3.6. The time complexity analysis of BPUSS-MOEA

According to the framework of BPUSS-MOEA shown in Algorithm 5, in this section, we present the time complexity analysis of BPUSSMOEA. Specifically, the whole process of the proposed BPUSS-MOEA is mainly composed of three steps.

The first step (Lines 1 - 2) is to generate the initial population and obtain the non-dominated individuals of the initial population. Specifically, the initial population is generated with the suggested dual-coding based population initialization strategy (see Algorithm 4), while the non-dominated sorting method suggested in NSGA-II is adopted to obtain the non-dominated individuals. Thus, their computational complexity is $\mathcal{O}(popsize \times m + popsize^2)$, where *popsize* is the size of the population and *m* is the length of the individual. The second step (Lines 3 - 9) is to evolve the population into the next generation. To be specific, the mating selection operation used in NSGA-II is performed, whose computational complexity is $\mathcal{O}(popsize^2)$. Then, the suggested dual-coding based offspring generation strategy (see Algorithm 1) is performed, which includes the binary coding part update (see Algorithm 2) and the real-valued coding part update (see Algorithm 3). Specifically, the computational complexity of the binary coding part update and the real-valued coding part update are both $\mathcal{O}(popsize \times m)$. Thus, the total computational complexity of Algorithm 1 is $\mathcal{O}(popsize \times m)$. Finally, we adopt the environmental selection method in NSGA-II to select the promising individuals into the next generation, whose computation cost is $\mathcal{O}(popsize * \log(popsize))$. The second step is repeated for *Maxiter* generations and the total computational complexity of the second step is $\mathcal{O}(Maxiter \times (popsize^2 + popsize \times m + popsize * \log(popsize)))$. The third step (Line 10) is to select the non-dominated individuals with the non-dominated sorting method, whose computation cost is $\mathcal{O}(popsize^2)$. According to the analysis above, the computational complexity of the proposed BPUSS-MOEA is $\mathcal{O}(popsize \times m + popsize^2 + Maxiter \times (popsize^2 + popsize \times m + popsize * \log(popsize)) + popsize^2)$, which is simplified as $\mathcal{O}(Maxiter \times (popsize^2 + popsize \times m))$.

4. Experimental results

In this section, a series of experiments is conducted to empirically verify the performance of proposed BPUSS-MOEA on 10 real-world datasets with different noise levels. Specifically, we firstly introduce the experimental settings, including the datasets, the compared algorithms and the parameter settings for all algorithms. Then, the comparison results are reported and analyzed in detail. Thirdly, the effectiveness of the proposed strategies (i.e., the dual-coding based offspring generation strategy and the dual-coding based population initialization strategy) in BPUSS-MOEA is validated. Finally, the parameter sensitivity analysis in BPUSS-MOEA is discussed.

4.1. Experimental settings

4.1.1. Datasets

To evaluate the performance of proposed BPUSS-MOEA, 10 real-world datasets are employed in the experiments, which have been widely used to validate the performance of the different PU algorithms [9,23,25,37]. These datasets can be downloaded from UCI machine learning repository.¹

Table 2 lists the detailed characteristics of 10 experimental datasets, where “#Samples” and “#Features” denote the number of samples and features, respectively. The number of positive samples and negative samples are respectively represented as “Num of P” and “Num of N”. To validate the performance of different algorithms, we introduce a certain percent noise into the original data.

¹ <https://archive.ics.uci.edu/ml/>.

To be specific, in this paper, we consider $n\%$ ($n = 0, 10, 20$) label noise level, which means for each dataset, we randomly change the labels of $n\%$ negative samples and add into P samples to generate the noisy P samples. Then, a widely used PU data generation approach is employed to construct the noisy PU datasets [23]. Specifically, the $\mu\%$ ($\mu = 20, 30, 40$) positive samples are randomly selected from the positive samples and together with the negative samples to consist of the U samples in PU dataset while the $(1-\mu\%)$ positive samples are treated as the P samples in PU dataset.

For the noisy PU dataset constructed by each pair of (n, μ) , five-fold cross validation is employed, where four folds are used for training the PU classifier and the remaining fold for testing. To further reduce the variance, each algorithm is independently run 30 times for further statistical analysis. The results averaging on 30 independent runs are reported, and the Wilcoxon rank-sum test with a significance level of 0.05 is adopted to perform statistical analysis.

4.1.2. Compared algorithms

The performance of the proposed BPUS-MOEA is compared with the four representative PU learning algorithms, namely, uPU [9], PULD [37], MOEA-PUL [23] and BAUC-OF [25]. To be specific, uPU and PULD are the PU learning algorithms that have shown the superior performance in the gradient based approaches. Specifically, in uPU, a ramp nonconvex loss function was employed to learn a PU classifier from the transformed cost-sensitive learning problem by estimating the class prior information. PULD is to treat each unlabeled sample as positive and negative class and the true label is identified by a suggested margin based label disambiguation strategy. MOEA-PUL is an MOEA-based PU algorithm, which is close to our algorithm. In MOEA-PUL, with the suggested single-coding scheme, the genetic operators are employed for identifying the labels of U samples. BAUC-OF is a recently suggested PU learning algorithm focusing on the PU learning with the noisy samples, which was tackled by the suggested outlier detection technique.

4.1.3. Parameter settings

For the uPU, PULD, MOEA-PUL and BAUC-OF algorithms, we adopt the recommended parameter values suggested in their original papers. Since the proposed BPUS-MOEA and MOEA-PUL both adopt the population based MOEA framework, for fair comparison, we fix the size of population ($popsize = 100$) and the maximum number of generation ($Maxiter = 100$) as the same with MOEA-PUL. In addition, in the proposed BPUS-MOEA, there is a parameter L used to control the size of neighbor, aiming to construct the C_{kNN} classifier. In the experiments, we set $L = 10$. For MOEA-PUL and the proposed BPUS-MOEA, the individual with the best classification performance is selected from the Pareto set (PS) and used as the final achieved PU classifier. All the experimental results are performed on a PC with an Intel Core i5-9500 3.0GHz CPU and Microsoft Windows 10 operating system.

4.2. Experimental results and analysis

4.2.1. The comparison results between BPUS-MOEA and baselines

In this section, the comparison results between BPUS-MOEA and four baselines are reported in terms of average classification accuracy, which are shown in Table 3 - Table 5. The highest accuracy on each dataset is marked in boldface, where $\mu\%$ is set to 20%, 30% and 40%, respectively. In addition, the last three lines in each table give the Wilcoxon significance statistics results, where the “win/tie/loss” denotes the results obtained by the proposed BPUS-MOEA are significantly better, similar and worse than those of the baselines.

Table 3 reports the average classification accuracy and the standard deviation of five PU learning algorithms on the PU data with different noise levels, when $\mu\% = 20\%$. From this table, we can find that the proposed BPUS-MOEA achieves the highest classification accuracy on 9/10 datasets with different label noises, which have shown the competitiveness of BPUS-MOEA. With increasing the level of noise, BPUS-MOEA has further shown the robustness compared with the other baselines in tackling the robust PU learning, which is verified by the Wilcoxon significance statistics. The superior performance of BPUS-MOEA is mainly attributed that it can select the “clean” PU samples and present the accurate labels of U samples to tackle the challenge of robust PU learning. In addition, when $n\% = 0$, that is no noisy samples in PU data, BPUS-MOEA still achieve the better or comparable performance with the other baselines.

Table 4 and Table 5 further present the average classification accuracy, when $\mu\% = 30\%$ and 40% . Although the increasing of U samples further add the challenge for the robust PU learning, BPUS-MOEA still obtain the competitive results compared to the other baselines, which is consistent with the results shown in Table 3. Specifically, compared with the uPU and PULD, with the increase of noise level, the BPUS-MOEA shows better performance, which is justified by the Wilcoxon significance test results. By further comparing BPUS-MOEA with MOEA-PUL, we can find that although they are both MOEAs for PU learning, the comparison results in Table 4 and Table 5 show that the average classification accuracy of MOEA-PUL is lower than that of BPUS-MOEA. The reason lies in the following fact. In MOEA-PUL, the adopted single-coding scheme is only used to represent the labels of U samples, which means that the noisy or redundant samples are all used to train the classifier. Thus, the performance of MOEA-PUL may be degraded. While in the proposed BPUS-MOEA, the designed dual-coding scheme can not only denote the candidate labels of U samples, but also select “clean” samples to construct the robust PU classifier, which ensures the superior performance of BPUS-MOEA under the noisy situation. In addition, despite BAUC-OF is designed to train the PU classifier in the context of the noisy positive samples, the statistics in Table 4 and Table 5 reveal that BPUS-MOEA always better or comparable to BAUC-OF in most cases. The reason is mainly attributed to the fact that BAUC-OF only consider the noisy samples in P samples. While in BPUS-MOEA, we consider not only the noises in P samples, but also the possible redundant samples in P and U samples, which ensures the performance of obtained PU classifier with higher quality.

Table 3

The average classification accuracy of BPUSS-MOEA vs. uPU, PULD, BAUC-OF and MOEA-PUL on the 10 datasets when $\mu\% = 20\%$.

Dataset	$n\%$	uPU	PULD	BAUC-OF	MOEA-PUL	BPUSS-MOEA
Sonar	0	0.606 ± 0.036	0.695 ± 0.059	0.683 ± 0.056	0.678 ± 0.037	0.89 ± 0.027
	10	0.485 ± 0.075	0.681 ± 0.049	0.674 ± 0.079	0.582 ± 0.081	0.852 ± 0.046
	20	0.471 ± 0.058	0.652 ± 0.043	0.683 ± 0.106	0.582 ± 0.081	0.838 ± 0.081
Heart	0	0.763 ± 0.036	0.833 ± 0.047	0.822 ± 0.065	0.844 ± 0.048	0.874 ± 0.053
	10	0.456 ± 0.092	0.848 ± 0.053	0.83 ± 0.036	0.459 ± 0.03	0.863 ± 0.036
	20	0.444 ± 0.023	0.826 ± 0.034	0.822 ± 0.017	0.47 ± 0.017	0.848 ± 0.051
Balance	0	0.944 ± 0.006	0.886 ± 0.036	0.907 ± 0.022	0.906 ± 0.025	0.918 ± 0.01
	10	0.461 ± 0.024	0.883 ± 0.046	0.912 ± 0.029	0.912 ± 0.016	0.93 ± 0.016
	20	0.462 ± 0.039	0.87 ± 0.056	0.912 ± 0.032	0.912 ± 0.016	0.917 ± 0.007
Breast	0	0.962 ± 0.021	0.739 ± 0.034	0.969 ± 0.019	0.972 ± 0.006	0.969 ± 0.016
	10	0.467 ± 0.271	0.965 ± 0.016	0.969 ± 0.012	0.365 ± 0.051	0.966 ± 0.02
	20	0.35 ± 0.046	0.959 ± 0.022	0.969 ± 0.010	0.365 ± 0.051	0.965 ± 0.016
Transfusion	0	0.703 ± 0.101	0.768 ± 0.022	0.622 ± 0.021	0.794 ± 0.024	0.815 ± 0.018
	10	0.239 ± 0.028	0.771 ± 0.024	0.620 ± 0.008	0.241 ± 0.034	0.776 ± 0.032
	20	0.235 ± 0.031	0.768 ± 0.028	0.618 ± 0.043	0.241 ± 0.034	0.785 ± 0.01
Diabetes	0	0.73 ± 0.041	0.753 ± 0.028	0.728 ± 0.018	0.784 ± 0.033	0.813 ± 0.029
	10	0.655 ± 0.034	0.736 ± 0.051	0.715 ± 0.044	0.655 ± 0.07	0.801 ± 0.016
	20	0.65 ± 0.036	0.764 ± 0.047	0.695 ± 0.024	0.655 ± 0.07	0.79 ± 0.042
Vehicle	0	0.65 ± 0.067	0.769 ± 0.008	0.654 ± 0.029	0.799 ± 0.029	0.808 ± 0.028
	10	0.252 ± 0.023	0.735 ± 0.037	0.652 ± 0.02	0.273 ± 0.032	0.809 ± 0.034
	20	0.255 ± 0.051	0.735 ± 0.037	0.636 ± 0.037	0.273 ± 0.032	0.824 ± 0.012
German	0	0.559 ± 0.03	0.761 ± 0.01	0.691 ± 0.043	0.774 ± 0.031	0.781 ± 0.034
	10	0.299 ± 0.027	0.732 ± 0.036	0.693 ± 0.034	0.327 ± 0.032	0.789 ± 0.012
	20	0.3 ± 0.036	0.73 ± 0.034	0.68 ± 0.037	0.327 ± 0.032	0.791 ± 0.019
Nba	0	0.695 ± 0.035	0.684 ± 0.021	0.646 ± 0.014	0.716 ± 0.032	0.721 ± 0.02
	10	0.616 ± 0.019	0.685 ± 0.024	0.651 ± 0.022	0.619 ± 0.041	0.723 ± 0.034
	20	0.619 ± 0.035	0.674 ± 0.014	0.647 ± 0.038	0.619 ± 0.041	0.727 ± 0.023
Madelon	0	0.5 ± 0.021	0.498 ± 0.01	0.588 ± 0.017	0.496 ± 0.025	0.689 ± 0.021
	10	0.499 ± 0.025	0.517 ± 0.025	0.584 ± 0.023	0.503 ± 0.023	0.679 ± 0.009
	20	0.5 ± 0.01	0.507 ± 0.017	0.58 ± 0.022	0.503 ± 0.023	0.656 ± 0.018
win/tie/loss	0	7/2/1	8/2/0	7/3/0	4/6/0	
	10	10/0/0	7/3/0	8/2/0	9/1/0	
	20	10/0/0	6/4/0	8/2/0	9/1/0	

The superiority of the proposed BPUSS-MOEA is that it can identify the “clean” PU samples for the robust PU learning. The selected PU samples can maintain the data distribution as the similar to that of the original PU samples. To verify this fact, we give the distribution of the original PU samples and the selected PU samples on the 10 datasets, where TSNE approach [29] is employed to project the datasets into the two-dimensional space for easy visualization. Due to the similarity of the results and the space limitation, three datasets (*Breast*, *Heart* and *Vehicle*) are randomly selected to verify the fact and the results are illustrated in Fig. 5, where $\mu\% = 30\%$ and $n\% = 10\%$. The left column and the right column of Fig. 5 respectively give the distribution of the original and the selected PU samples. As can be seen from these figures, the number of the selected PU samples are much smaller than that of the original PU samples but can maintain the similar sample distribution as the original ones. The results further demonstrate the fact that selecting the “clean” samples from the noisy and redundant PU data can be used to construct the robust PU classifier with high quality.

4.2.2. The effectiveness of the suggested strategies in BPUSS-MOEA

The reason that the proposed BPUSS-MOEA can obtain more robust PU classifier mainly attributed to the two strategies suggested in BPUSS-MOEA. Thus, in this section, we will verify the effectiveness of these two strategies, which are the dual-coding based offspring generation strategy and the dual-coding based population initialization strategy, respectively.

Table 4

The average classification accuracy of BPUS-MOEA vs. uPU, PULD, BAUC-OF and MOEA-PUL on the 10 datasets when $\mu\% = 30\%$.

Dataset	$n\%$	uPU	PULD	BAUC-OF	MOEA-PUL	BPUS-MOEA
Sonar	0	0.582 ± 0.062	0.671 ± 0.043	0.654 ± 0.055	0.683 ± 0.061	0.848 ± 0.064
	10	0.467 ± 0.093	0.652 ± 0.036	0.682 ± 0.055	0.495 ± 0.071	0.8 ± 0.013
	20	0.467 ± 0.092	0.648 ± 0.049	0.73 ± 0.096	0.495 ± 0.071	0.857 ± 0.041
Heart	0	0.781 ± 0.062	0.822 ± 0.034	0.822 ± 0.07	0.819 ± 0.051	0.87 ± 0.075
	10	0.493 ± 0.034	0.826 ± 0.036	0.844 ± 0.025	0.496 ± 0.133	0.859 ± 0.068
	20	0.452 ± 0.056	0.819 ± 0.048	0.833 ± 0.059	0.5 ± 0.068	0.841 ± 0.043
Balance	0	0.93 ± 0.018	0.869 ± 0.023	0.91 ± 0.027	0.922 ± 0.015	0.928 ± 0.015
	10	0.461 ± 0.031	0.84 ± 0.037	0.914 ± 0.017	0.91 ± 0.018	0.925 ± 0.012
	20	0.467 ± 0.029	0.781 ± 0.077	0.91 ± 0.01	0.91 ± 0.018	0.939 ± 0.011
Breast	0	0.962 ± 0.012	0.953 ± 0.02	0.968 ± 0.022	0.975 ± 0.012	0.966 ± 0.015
	10	0.35 ± 0.049	0.959 ± 0.017	0.968 ± 0.014	0.579 ± 0.333	0.966 ± 0.022
	20	0.483 ± 0.259	0.964 ± 0.018	0.968 ± 0.007	0.579 ± 0.333	0.958 ± 0.022
Transfusion	0	0.741 ± 0.069	0.769 ± 0.024	0.648 ± 0.017	0.794 ± 0.04	0.785 ± 0.012
	10	0.239 ± 0.035	0.765 ± 0.027	0.652 ± 0.03	0.241 ± 0.025	0.775 ± 0.032
	20	0.238 ± 0.022	0.756 ± 0.044	0.602 ± 0.046	0.241 ± 0.025	0.807 ± 0.005
Diabetes	0	0.753 ± 0.038	0.745 ± 0.056	0.717 ± 0.023	0.788 ± 0.043	0.758 ± 0.022
	10	0.651 ± 0.045	0.758 ± 0.034	0.726 ± 0.042	0.659 ± 0.048	0.799 ± 0.018
	20	0.652 ± 0.041	0.745 ± 0.039	0.71 ± 0.026	0.659 ± 0.048	0.786 ± 0.027
Vehicle	0	0.681 ± 0.06	0.754 ± 0.033	0.655 ± 0.024	0.793 ± 0.017	0.825 ± 0.022
	10	0.267 ± 0.056	0.754 ± 0.033	0.655 ± 0.035	0.306 ± 0.063	0.807 ± 0.038
	20	0.254 ± 0.042	0.754 ± 0.033	0.667 ± 0.068	0.306 ± 0.063	0.794 ± 0.039
German	0	0.567 ± 0.022	0.745 ± 0.026	0.681 ± 0.032	0.769 ± 0.029	0.795 ± 0.038
	10	0.315 ± 0.051	0.718 ± 0.006	0.681 ± 0.031	0.363 ± 0.037	0.778 ± 0.029
	20	0.302 ± 0.037	0.677 ± 0.039	0.683 ± 0.039	0.363 ± 0.037	0.751 ± 0.024
Nba	0	0.681 ± 0.018	0.674 ± 0.012	0.647 ± 0.026	0.715 ± 0.028	0.691 ± 0.022
	10	0.62 ± 0.061	0.651 ± 0.011	0.649 ± 0.042	0.617 ± 0.04	0.686 ± 0.037
	20	0.62 ± 0.049	0.651 ± 0.011	0.646 ± 0.037	0.617 ± 0.04	0.713 ± 0.034
Madelon	0	0.5 ± 0.028	0.497 ± 0.011	0.591 ± 0.023	0.617 ± 0.031	0.692 ± 0.017
	10	0.5 ± 0.031	0.497 ± 0.004	0.586 ± 0.023	0.52 ± 0.019	0.665 ± 0.016
	20	0.5 ± 0.016	0.508 ± 0.026	0.585 ± 0.024	0.52 ± 0.019	0.645 ± 0.021
win/tie/loss	0	5/5/0	6/4/0	7/3/0	4/6/0	
	10	10/0/0	7/3/0	6/4/0	9/1/0	
	20	10/0/0	8/2/0	8/2/0	10/0/0	

Table 6 gives the average classification accuracy obtained by the proposed BPUS-MOEA and its variant BPUS-MOEA-WDO. Specifically, in BPUS-MOEA-WDO, all components keep the same as the BPUS-MOEA except that the suggested dual-coding based offspring generation strategy is replaced by the crossover and mutation strategies suggested in NSGA-II [8]. Note that “WDO” denotes “without the dual-coding offspring generation strategy”. Due to the similarity of the results and the space limitation, in the following experiments, we only present the results when $\mu\% = 30\%$. From this table, we can find that BPUS-MOEA outperforms the BPUS-MOEA-WDO in terms of the average accuracy on all datasets. Since canceling the dual-coding based offspring generation strategy, BPUS-MOEA-WDO obtains the offspring individuals by using the crossover and mutation operators employed in NSGA-II [8], which results in lower classification accuracy. The reason is mainly attributed to the following fact. The crossover and mutation operators in NSGA-II designed for the single-coding individuals can not identify and remove the possible noisy and redundant PU samples, which results in the difficulty of obtaining the robust PU classifier.

Table 7 presents the average classification accuracy of the proposed BPUS-MOEA and its variant BPUS-MOEA-RI. In BPUS-MOEA-RI, the suggested dual-coding based population initialization strategy is replaced by random initialization strategy and all the other components keep the same as the BPUS-MOEA. Note that “RI” denotes “random initialization strategy”. As shown in this table, we can find that the PU classifiers obtained by BPUS-MOEA outperform the variant BPUS-MOEA-RI in terms of the average test accuracy. To be specific, due to the designed dual-coding initialization strategy, the PU classifier constructed by the selected

Table 5

The average classification accuracy of BPUS-MOEA vs. uPU, PULD, BAUC-OF and MOEA-PUL on the 10 datasets when $\mu\% = 40\%$.

Dataset	$n\%$	uPU	PULD	BAUC-OF	MOEA-PUL	BPUS-MOEA
Sonar	0	0.592 ± 0.116	0.662 ± 0.043	0.692 ± 0.094	0.735 ± 0.031	0.805 ± 0.031
	10	0.466 ± 0.053	0.614 ± 0.085	0.693 ± 0.142	0.485 ± 0.134	0.781 ± 0.046
	20	0.471 ± 0.056	0.576 ± 0.035	0.726 ± 0.078	0.543 ± 0.105	0.781 ± 0.054
Heart	0	0.763 ± 0.097	0.833 ± 0.035	0.826 ± 0.021	0.83 ± 0.03	0.837 ± 0.04
	10	0.444 ± 0.064	0.811 ± 0.084	0.819 ± 0.078	0.481 ± 0.1	0.822 ± 0.062
	20	0.448 ± 0.05	0.781 ± 0.024	0.822 ± 0.046	0.596 ± 0.184	0.822 ± 0.056
Balance	0	0.938 ± 0.021	0.845 ± 0.031	0.914 ± 0.02	0.922 ± 0.024	0.925 ± 0.012
	10	0.501 ± 0.087	0.763 ± 0.078	0.92 ± 0.022	0.904 ± 0.02	0.934 ± 0.025
	20	0.461 ± 0.047	0.491 ± 0.086	0.914 ± 0.025	0.909 ± 0.036	0.906 ± 0.007
Breast	0	0.952 ± 0.018	0.961 ± 0.012	0.969 ± 0.014	0.963 ± 0.021	0.942 ± 0.036
	10	0.48 ± 0.28	0.965 ± 0.02	0.969 ± 0.023	0.534 ± 0.206	0.933 ± 0.022
	20	0.36 ± 0.038	0.943 ± 0.024	0.968 ± 0.008	0.581 ± 0.321	0.926 ± 0.049
Transfusion	0	0.679 ± 0.086	0.765 ± 0.028	0.646 ± 0.021	0.787 ± 0.03	0.773 ± 0.029
	10	0.238 ± 0.034	0.757 ± 0.043	0.61 ± 0.032	0.27 ± 0.067	0.787 ± 0.051
	20	0.239 ± 0.04	0.693 ± 0.041	0.632 ± 0.027	0.266 ± 0.055	0.78 ± 0.03
Diabetes	0	0.745 ± 0.065	0.734 ± 0.056	0.717 ± 0.029	0.793 ± 0.034	0.771 ± 0.01
	10	0.654 ± 0.021	0.751 ± 0.057	0.721 ± 0.015	0.671 ± 0.046	0.735 ± 0.045
	20	0.65 ± 0.033	0.717 ± 0.057	0.699 ± 0.047	0.647 ± 0.045	0.734 ± 0.038
Vehicle	0	0.677 ± 0.048	0.729 ± 0.053	0.649 ± 0.01	0.8 ± 0.04	0.764 ± 0.044
	10	0.252 ± 0.026	0.729 ± 0.053	0.654 ± 0.03	0.409 ± 0.096	0.78 ± 0.026
	20	0.261 ± 0.04	0.729 ± 0.053	0.648 ± 0.081	0.348 ± 0.093	0.761 ± 0.038
German	0	0.534 ± 0.079	0.735 ± 0.02	0.677 ± 0.023	0.776 ± 0.026	0.757 ± 0.027
	10	0.3 ± 0.013	0.681 ± 0.038	0.679 ± 0.015	0.476 ± 0.114	0.758 ± 0.029
	20	0.3 ± 0.062	0.643 ± 0.048	0.673 ± 0.032	0.436 ± 0.086	0.765 ± 0.028
Nba	0	0.693 ± 0.019	0.648 ± 0.037	0.651 ± 0.02	0.702 ± 0.047	0.607 ± 0.029
	10	0.619 ± 0.029	0.602 ± 0.014	0.649 ± 0.05	0.608 ± 0.024	0.625 ± 0.028
	20	0.619 ± 0.028	0.602 ± 0.014	0.648 ± 0.04	0.616 ± 0.019	0.662 ± 0.028
Madelon	0	0.5 ± 0.011	0.502 ± 0.019	0.582 ± 0.017	0.616 ± 0.031	0.666 ± 0.017
	10	0.501 ± 0.016	0.507 ± 0.017	0.574 ± 0.022	0.48 ± 0.035	0.673 ± 0.023
	20	0.499 ± 0.017	0.506 ± 0.023	0.563 ± 0.031	0.506 ± 0.017	0.628 ± 0.021
win/tie/loss	0	5/4/1	3/6/1	6/3/1	3/6/1	
	10	9/1/0	6/3/1	4/5/1	9/1/0	
	20	10/0/0	6/4/0	4/5/1	9/1/0	

PU samples has higher classification accuracy. The reason is attributed that in the suggested initialization strategy, we construct a classifier C_{kNN} to obtain the predicted labels of the neighbors of each sample, with which to obtain the initial individuals with high qualities. Thus, the effectiveness of the proposed strategies used in BPUS-MOEA is verified.

4.2.3. Sensitivity analysis on the parameter L in BPUS-MOEA

In the proposed BPUS-MOEA, the parameter L is used to control the number of the neighbors of each sample, which is key to the proposed BPUS-MOEA. Thus, in this section, we empirically investigate the influence of parameter L on 10 datasets. The results of the accuracy on each dataset with different L are illustrated in Fig. 6, where L is selected from 1 to 30 with the interval of 5. As can be seen from this figure, when the value of the parameter L set small (e.g., $L = 1$), the performance of the obtained PU classifier is not good. The reason is that when the value of L is too small, the effective neighboring information is not enough to predict the accurate labels, which may degrade the performance of the PU classifiers. When the parameter is set large (e.g., $L = 25$), the performance of BPUS-MOEA is also not the optimal. The reason lies in the fact that with the increasing of L , more neighbor samples are used to construct the C_{kNN} classifier, which includes the possible noisy and redundant samples and influences the performance of the proposed BPUS-MOEA. Thus, from this figure, it is appropriate for the parameter L when its value is set between 10 and 15. Thus, in all the experiments, we empirically set the parameter $L = 10$.

Table 6

The average classification accuracy obtained by BPUSS-MOEA-WDO and BPUSS-MOEA on the 10 datasets when $\mu\% = 30\%$.

Dataset	$n\%$	BPUSS-MOEA-WDO	BPUSS-MOEA
Sonar	0	0.759 ± 0.047	0.848 ± 0.064
	10	0.759 ± 0.07	0.800 ± 0.013
	20	0.721 ± 0.042	0.857 ± 0.041
Heart	0	0.807 ± 0.028	0.87 ± 0.075
	10	0.837 ± 0.053	0.859 ± 0.068
	20	0.815 ± 0.08	0.841 ± 0.043
Balance	0	0.901 ± 0.009	0.928 ± 0.015
	10	0.918 ± 0.026	0.925 ± 0.012
	20	0.91 ± 0.033	0.939 ± 0.011
Breast	0	0.946 ± 0.008	0.966 ± 0.015
	10	0.95 ± 0.022	0.966 ± 0.022
	20	0.953 ± 0.015	0.958 ± 0.022
Transfusion	0	0.774 ± 0.033	0.785 ± 0.012
	10	0.789 ± 0.024	0.775 ± 0.032
	20	0.79 ± 0.019	0.807 ± 0.005
Diabetes	0	0.772 ± 0.021	0.758 ± 0.022
	10	0.753 ± 0.059	0.799 ± 0.018
	20	0.768 ± 0.02	0.786 ± 0.027
Vehicle	0	0.777 ± 0.031	0.825 ± 0.022
	10	0.783 ± 0.03	0.807 ± 0.038
	20	0.773 ± 0.034	0.794 ± 0.039
German	0	0.752 ± 0.029	0.795 ± 0.038
	10	0.755 ± 0.018	0.778 ± 0.029
	20	0.746 ± 0.019	0.751 ± 0.024
Nba	0	0.663 ± 0.038	0.691 ± 0.022
	10	0.684 ± 0.025	0.686 ± 0.037
	20	0.696 ± 0.029	0.713 ± 0.034
Madelon	0	0.552 ± 0.029	0.692 ± 0.017
	10	0.543 ± 0.028	0.665 ± 0.016
	20	0.55 ± 0.008	0.645 ± 0.021
win/tie/loss	0	7/3/0	
	10	5/5/0	
	20	5/5/0	

5. Conclusions and future work

In this paper, the robust PU learning was transformed into a bi-objective PU samples selection (BPUSS) problem. Specifically, the first objective was to measure the ratio of selected “clean” PU samples to all the PU samples. The second objective was designed to measure the accuracy of PU classifier trained on the selected “clean” PU samples. To tackle the BPUSS problem, a novel MOEA, named BPUSS-MOEA, was proposed. In BPUSS-MOEA, we designed a dual-coding scheme with the binary coding and the real-valued coding. The binary coding was used to determine whether a PU sample is selected as a “clean” sample. The real-valued coding was employed to denote the labels of U samples and the selection probability of each PU sample. With the suggested dual-coding scheme, a two-stage updating process was proposed to generate the promising offspring individuals. To be specific, the first stage was to update the binary coding part. The second stage was to update the real-valued coding part. Additionally, to further enhance the performance of the proposed BPUSS-MOEA, a dual-coding base population initialization strategy was suggested to obtain the promising initial population. Experimental results have demonstrated the performance of the BPUSS-MOEA on 10 PU datasets with different noise levels, compared with the state-of-the-arts.

In this work, we mainly focus on robust PU learning from the noisy and redundant PU data. The proposed BPUSS-MOEA has shown the promising performance in our experiments. In BPUSS-MOEA, we employ the population-based MOEA framework, where

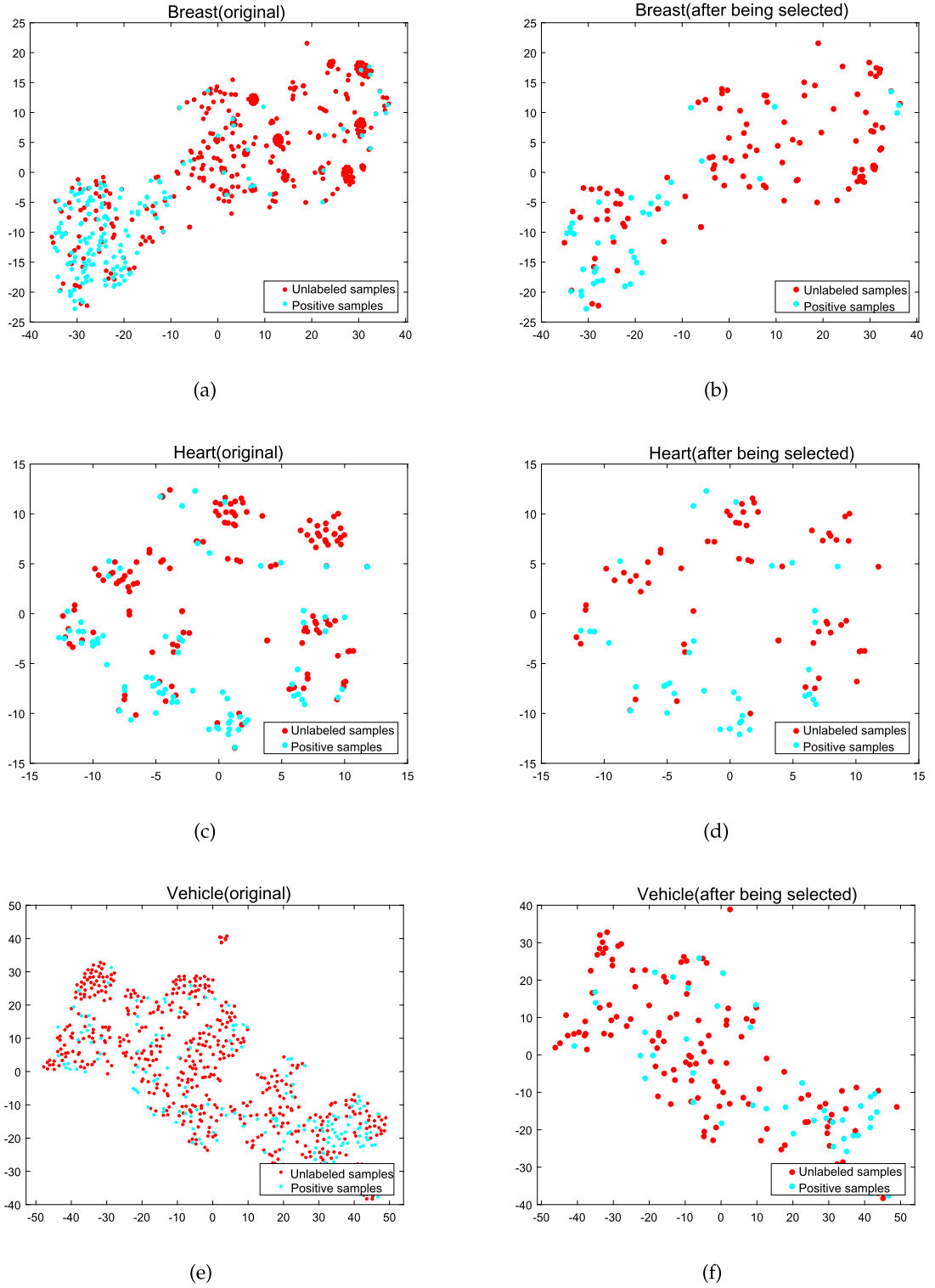


Fig. 5. The distribution of the original PU samples and the selected PU samples obtained by the proposed BPUS-MOEA, where $\mu\% = 30\%$ and $n\% = 10\%$.

the length of each individual is equal to the number of PU samples. As the size of PU data increases, the computing efficiency of proposed algorithm will decrease, which will be further considered in the future work. Thus, how to design more efficient PU algorithms for robust PU learning with a great deal of data is our future work.

Table 7

The average classification accuracy obtained by BPUSS-MOEA-RI and BPUSS-MOEA on the 10 datasets when $\mu\% = 30\%$.

Dataset	$n\%$	BPUSS-MOEA-RI	BPUSS-MOEA
Sonar	0	0.706 ± 0.073	0.848 ± 0.064
	10	0.707 ± 0.064	0.800 ± 0.013
	20	0.657 ± 0.071	0.857 ± 0.041
Heart	0	0.777 ± 0.026	0.87 ± 0.075
	10	0.679 ± 0.017	0.859 ± 0.068
	20	0.786 ± 0.018	0.841 ± 0.043
Balance	0	0.907 ± 0.02	0.928 ± 0.015
	10	0.847 ± 0.011	0.925 ± 0.012
	20	0.875 ± 0.018	0.939 ± 0.011
Breast	0	0.89 ± 0.008	0.966 ± 0.015
	10	0.897 ± 0.009	0.966 ± 0.022
	20	0.924 ± 0.003	0.958 ± 0.022
Transfusion	0	0.672 ± 0.091	0.785 ± 0.012
	10	0.634 ± 0.083	0.775 ± 0.032
	20	0.581 ± 0.107	0.807 ± 0.005
Diabetes	0	0.746 ± 0.053	0.758 ± 0.022
	10	0.645 ± 0.041	0.799 ± 0.018
	20	0.668 ± 0.052	0.786 ± 0.027
Vehicle	0	0.657 ± 0.048	0.825 ± 0.022
	10	0.707 ± 0.048	0.807 ± 0.038
	20	0.721 ± 0.055	0.794 ± 0.039
German	0	0.648 ± 0.055	0.795 ± 0.038
	10	0.519 ± 0.051	0.778 ± 0.029
	20	0.608 ± 0.051	0.751 ± 0.024
Nba	0	0.595 ± 0.035	0.691 ± 0.022
	10	0.51 ± 0.029	0.686 ± 0.037
	20	0.514 ± 0.033	0.713 ± 0.034
Madelon	0	0.545 ± 0.017	0.692 ± 0.017
	10	0.414 ± 0.015	0.665 ± 0.016
	20	0.504 ± 0.029	0.645 ± 0.021
win/tie/loss	0	9/1/0	
	10	10/0/0	
	20	8/2/0	

CRedit authorship contribution statement

Jianfeng Qiu: Writing – review & editing, Writing – original draft. **Qi Tang:** Writing – original draft, Writing – review & editing. **Ming Tan:** Writing – review & editing. **Kaixuan Li:** Writing – review & editing. **Juan Xie:** Writing – review & editing. **Xiaoqiang Cai:** Writing – review & editing. **Fan Cheng:** Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

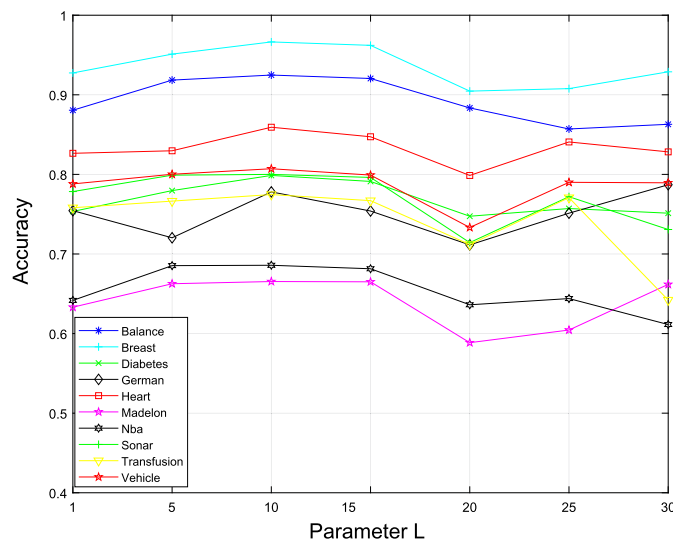


Fig. 6. The sensitivity analysis of the parameter L , when $\mu\% = 30\%$ and $n\% = 10\%$.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ins.2024.120992>.

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